

Advanced Data Analytics — Lecture 9

Deep Learning for Solving Dynamic Models in Economics and Finance

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Summary

Motivation: High-Dimensional Dynamic Models

- ▶ Modern macro/finance: heterogeneity, aggregate shocks, occasionally binding constraints
- ▶ Need global policy functions on irregular ergodic sets (grid-free preferred)
- ▶ **DEQNs**: learn equilibrium objects by minimizing economic residuals (Euler, constraints)

Curse of dimensionality: volume of sphere vs. cube $\rightarrow 0$ as $d \rightarrow \infty$

Neural Networks as Function Approximators

$$\mathcal{N}_\rho(x) = \sigma_K(W_K \dots \sigma_1(W_1 x + b_1) \dots + b_K)$$

- ▶ Universal approximators; train with (stochastic) gradient descent
- ▶ Use as *policy function emulator* $k' = f_\rho(k, z)$

Deep Equilibrium Nets (DEQNs): Key Idea

Unsupervised, economics-informed training

Minimize the squared residuals of equilibrium conditions (Euler, feasibility, etc.):

$$\mathcal{L}(\rho) = \mathbb{E} \left[\sum_{j \in \text{batch}} (\text{EulerResidual}_j(\rho))^2 \right]$$

- ▶ No labels required; states sampled from simulated paths
- ▶ Specialize here to simple growth models (with/without shocks)

Planner's Problem (Deterministic)

Technology: $y = f(k) = Ak^\alpha$, $A > 0$, $\alpha \in (0, 1)$

Resource constraint: $c_t + k_{t+1} = f(k_t)$

Preferences: $\max_{\{c_t\}_{t \geq 0}} \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma}$, $\beta \in (0, 1)$, $\sigma > 0$

Bellman equation:

$$V(k) = \max_{k' \in [0, f(k)]} \underbrace{\frac{(f(k) - k')^{1-\sigma}}{1-\sigma}}_{u(c)} + \beta V(k')$$

with $c = f(k) - k'$.

FOCs, Envelope, Euler, TVC

FOC wrt k' (via $c = f(k) - k'$):

$$\frac{\partial}{\partial k'} \left(u(c) + \beta V(k') \right) = -u'(c) + \beta V'(k') = 0 \Rightarrow u'(c) = \beta V'(k')$$

Envelope Theorem:

$$V(k) = u'(c) f'(k)$$

Euler equation (combine FOC at t with envelope at $t+1$):

$$u'(c_t) = \beta V'(k_{t+1}) = \beta u'(c_{t+1}) f'(k_{t+1})$$

For CRRA $u'(c) = c^{-\sigma}$ and $f'(k) = \alpha A k^{\alpha-1}$:

$$c_t^{-\sigma} = \beta c_{t+1}^{-\sigma} \alpha A k_{t+1}^{\alpha-1}.$$

Transversality condition (TVC):

$$\lim_{T \rightarrow \infty} \beta^T u'(c_T) k_{T+1} = 0.$$

Closed-Form Policy and Steady State

For CRRA and Cobb–Douglas (Brock–Mirman):

$$\boxed{k_{t+1} = \alpha\beta A k_t^\alpha}, \quad \boxed{c_t = (1 - \alpha\beta) A k_t^\alpha}.$$

Steady state k^* :

$$k^* = (\alpha\beta A)^{\frac{1}{1-\alpha}}, \quad c^* = (1 - \alpha\beta) A (k^*)^\alpha.$$

Relative Euler error (training target):

$$e^{\text{REE}}(k; \rho) = \frac{u'(c(k; \rho))}{\beta u'(c(k'; \rho)) f'(k')} - 1 \quad \text{with} \quad k' = f_\rho(k).$$

Loss (deterministic):
$$\mathcal{L}_{\text{det}}(\rho) = \frac{1}{|\mathcal{D}_{\text{train}}|} \sum_{k \in \mathcal{D}_{\text{train}}} (e^{\text{REE}}(k; \rho))^2.$$

Stochastic Environment

Technology with shock: $y_t = z_t A k_t^\alpha$, $z_{t+1} \sim \mathcal{P}(\cdot | z_t)$

$$c_t + k_{t+1} = z_t A k_t^\alpha$$

Bellman equation:

$$V(k, z) = \max_{k' \in [0, z A k^\alpha]} \frac{(z A k^\alpha - k')^{1-\sigma}}{1-\sigma} + \beta \mathbb{E} [V(k', z') | z].$$

FOC (using $c = z A k^\alpha - k'$):

$$-u'(c) + \beta \mathbb{E} [V_k(k', z') | z] = 0 \quad \Rightarrow \quad u'(c_t) = \beta \mathbb{E} [V_k(k_{t+1}, z_{t+1})].$$

Envelope & Euler with Expectation

Envelope Theorem:

$$V_k(k, z) = u'(c) \frac{\partial c}{\partial k} = u'(c) z A \alpha k^{\alpha-1} = u'(c) f_k(k, z).$$

Euler equation (substitute envelope into FOC):

$$u'(c_t) = \beta E [{}_t [u'(c_{t+1}) f_k(k_{t+1}, z_{t+1})]]$$

i.e.

$$c_t^{-\sigma} = \beta E [{}_t [c_{t+1}^{-\sigma} \alpha A z_{t+1} k_{t+1}^{\alpha-1}]].$$

TVC (stochastic): $\lim_{T \rightarrow \infty} \beta^T E [{}_0 [u'(c_T) k_{T+1}]] = 0.$

Residuals and DEQN Loss (Stochastic)

Let $k' = f_\rho(k, z)$, $c = zAk^\alpha - k'$, and draw $z'|z$ from $\mathcal{P}$.

$$e^{\text{REE}}(k, z; \rho) = \frac{u'(c)}{\beta \mathbb{E} [u'(c') f_k(k', z') \mid z]} - 1.$$

Mini-batch loss:

$$\mathcal{L}_{\text{sto}}(\rho) = \frac{1}{|\text{batch}|} \sum_{(k,z) \in \text{batch}} \left(e^{\text{REE}}(k, z; \rho) \right)^2,$$

with the expectation computed *exactly* for discrete z' or via quadrature/Monte Carlo for continuous z' .

Feasibility projection (if needed early in training): $k' \in [0, zAk^\alpha]$; penalize $c < 0$ heavily.

Economics-Informed Training Loop

1. Initialize ρ ; choose batch generator for (k) or (k, z) along simulated paths.
2. For each batch:
 - ▶ Compute $k' = f_\rho(\cdot)$, c , c' , $f_k(\cdot)$.
 - ▶ Form e^{REE} (deterministic or stochastic).
 - ▶ Update $\rho \leftarrow \rho - \alpha_{\text{learn}} \nabla_\rho \mathcal{L}$.
3. Repeat (episodes) with refreshed simulated states $\mathcal{D}_{\text{train}}$.

Note: No parallelization/OLG content here; loss involves only Euler residuals of the simple models.

Notebook 1: No Uncertainty & Exogenous Sampling

- ▶ States: sample k over relevant ergodic range
- ▶ Target residual: deterministic e^{REE}
- ▶ Diagnostics: plot cost decay; check max and percentiles of REE
- ▶ Sanity checks: implied k' and c near closed-form $(\alpha\beta A k^\alpha, (1 - \alpha\beta) A k^\alpha)$

Notebook 2: With Uncertainty (Discrete z)

- ▶ State: (k, z) ; simulate z via Markov chain
- ▶ Expectation $E[t[\cdot]]$: exact sum over z' (or MC if continuous)
- ▶ Residual: stochastic $e^{\text{REE}}(k, z)$
- ▶ Diagnostics: REE by state; stability of feasible set ($c \geq 0$)

Summary

- ▶ Two Brock–Mirman models (deterministic & stochastic) derived from first principles
- ▶ **Envelope Theorem** $\Rightarrow$ Euler equation (with/without expectation)
- ▶ DEQN trains on *relative Euler residuals* — no labels, grid-free
- ▶ Deterministic case: closed-form benchmark for validation; stochastic case: exact expectation for discrete shocks

Next: plug in your notebook plots to the placeholders.